FinMini: Two-Stage Fine-Tuning of a Compact LLM for Financial Sentiment Analysis and Text-Driven Stock Movement Prediction

JiaYi Zhang, JiaYuan Jiang, Yue Yu, YiXiang Fei

jiayizha@umich.edu, jiayuanj@umich.edu, yuyuea@umich.edu, feiyx@umich.edu

Electrical Engineering and Computer Science, University of Michigan

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Abstract

Large language models (LLMs) excel at general-purpose tasks yet consistently underperform in specialized financial domains, where domain-specific terminology, implicit sentiment cues, and causal reasoning differ fundamentally from everyday language. We present **FinMini**, a two-stage domain adaptation pipeline applied to the compact 8B-parameter model *DeepSeek-R1-Distill-Llama-8B*. The first stage performs quantized LoRA supervised fine-tuning (QLoRA-SFT) on 63,395 labeled financial samples drawn from three curated corpora. The second stage extends training with Chain-of-Thought SFT (CoT-SFT), replacing bare sentiment labels with structured `<think>` reasoning traces generated via the DeepSeek API, aligned with the base model's native format. Evaluated on four benchmarks from the Open FinLLM Leaderboard — FPB, FiQASA, SM-BigData, and SM-CIKM — FinMini CoT-SFT achieves **98.5%**, **97.7%**, **57.3%**, and **55.5%** accuracy respectively, outperforming all evaluated baselines including Mistral-7B-Instruct and Meta-Llama-3-8B-Instruct. We further demonstrate an end-to-end application by deploying the fine-tuned model in a quarterly long-short equity strategy over the S&P 500 universe, showing that LLM-derived sentiment scores can serve as meaningful cross-sectional return signals.

1 Introduction

Financial natural language processing (NLP) is challenging because earnings calls, analyst reports, news, and social media contain domain-specific vocabulary and implicit market expectations [4]. Lexicon-based methods such as the Loughran–McDonald dictionary [6] address financial vocabulary but cannot capture context-dependent sentiment.

We ask whether a compact open-source LLM can be efficiently adapted for financial sentiment analysis and stock movement prediction. Both tasks are framed as binary positive/negative classification following the Open FinLLM Leaderboard [5].

Our main contributions are:

1. A **two-stage fine-tuning pipeline** combining QLoRA-SFT and CoT-SFT for a compact reasoning-distilled LLM.
2. A **CoT data augmentation strategy** that leverages the DeepSeek API to synthesize structured `<think>` reasoning traces for over 60,000 financial samples.
3. Comprehensive **benchmarking** of three open-source 7–8B models, with staged ablation across fine-tuning phases.

4. An **end-to-end trading strategy demonstration** that applies FinMini to S&P 500 earnings call scoring and long-short portfolio construction.

2 Related Work

Early financial sentiment systems relied on domain lexicons such as the Loughran–McDonald dictionary [6] and classical bag-of-words classifiers [2]. These methods address financial vocabulary but cannot fully model context or causal language. FinBERT [1] improved performance through domain-specific pretraining, but encoder-only models lack the instruction-following and reasoning behavior of modern decoder-only LLMs.

Recent financial LLM work includes BloombergGPT [9], which relies on large proprietary pretraining, and PIXIU/FinBen [10], which shows the value of financial instruction tuning. Our work follows the Open FinLLM evaluation setting and combines FPB with FinGPT/Twitter data [8] and stock movement tasks. Long CoT SFT [7] further motivates our second stage: we use supervised `<think>` traces to strengthen reasoning without requiring reinforcement learning.

3 Method

3.1 Base Model Selection

We select **DeepSeek-R1-Distill-Llama-8B** as our base model for three reasons. First, it incorporates reasoning-distillation pretraining, which endows it with stronger structured reasoning out of the box compared with standard instruction-tuned models. Second, among zero-shot baselines (Section 4), it achieves the highest average accuracy on social-media benchmarks, suggesting a better-initialized prior for ambiguous financial text. Third, its native `<think>`...`</think>` output format is directly compatible with our CoT-SFT augmentation strategy.

3.2 QLoRA Supervised Fine-Tuning

We apply **QLoRA** [3] to reduce memory footprint while preserving model quality. The base model is loaded in 4-bit NF4 double quantization (bitsandbytes), keeping all original weights frozen. Low-rank adapter matrices are injected into all seven projection matrices: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`. Formally, for a pretrained weight matrix $W_0 \in \mathbb{R}^{d\times k}$, the adapted forward pass is:

$$h = W_0 x + \frac{\alpha}{r} BAx, \quad A \in \mathbb{R}^{r\times k},\ B \in \mathbb{R}^{d\times r},$$

where rank $r = 16$ and scaling factor $\alpha = 32$. Only A and B are updated during training, reducing trainable parameters from $\sim$8B to $\sim$42M while maintaining full model expressiveness via the quantized base.

Training configuration. We train for 3 epochs with per-device batch size 4 and gradient accumulation of 4 steps (effective batch size 16), using paged AdamW at learning rate 2×10^{-4} with a cosine decay schedule. Maximum sequence length is 512 tokens with BF16 mixed precision.

3.3 Training Corpus

Each sample follows an instruction/input/output template. The instruction describes the task; the input contains the financial text; the output is the sentiment label (`positive` / `negative`).

Dataset	Samples	Selection Rationale
FPB (Financial PhraseBank)	2,264	100% inter-annotator agreement; gold-standard annotation quality
DS2 (FinGPT + Twitter [8])	100,037	Multi-source scale; covers news, social media, and analyst commentary
FLARE-SM-ACL	27,056	Stock movement reasoning; extends task diversity beyond pure sentiment
Total (train / test split)	**63,395 / 7,979**	

Table 1: Training corpus. Positive/negative class distribution: 41,211 / 22,184 in train; 5,235 / 2,744 in test. Neutral labels are removed to unify the binary classification framework.

Proportional sampling is applied so that the large DS2 corpus does not overwhelm the high-quality FPB annotations.

3.4 Chain-of-Thought Supervised Fine-Tuning

The first-stage adapter is retained and training continues on a CoT-augmented version of the corpus. For each training sample, we query **DeepSeek-chat** (temperature 0.7, `max_tokens`= 300) to generate a structured reasoning chain. The output format is transformed from a bare label into:

`<think>` [financial reasoning] `</think>` `positive` | `negative`

Training uses `completion_only_loss`: only the completion tokens — the `<think>` trace plus the final label — contribute to the gradient, not the prompt. Maximum sequence length is extended to 2,048 tokens to accommodate reasoning traces. The CoT augmentation was applied to the full Stage 1 training set (63,395 samples); after filtering out samples where the API call failed or the output did not contain a valid `<think>` block, and applying a 90/10 split, 54,595 / 6,067 samples are used for this stage.

Figure 1 shows the training loss curves for both stages. The first stage converges from ∼4.0 to ∼1.8; the second stage converges from ∼1.0 to ∼0.5, reflecting the richer supervision signal provided by reasoning traces compared with label-only outputs.

4 Experiments

4.1 Evaluation Benchmarks

We evaluate on four benchmarks from the Open FinLLM Leaderboard [5]:

All models use 4-bit NF4 quantization, `max_new_tokens`= 32, and greedy decoding. For models producing `<think>`…`</think>` blocks, the block is stripped before label extraction.

4.2 Baselines

Three open-source models are evaluated zero-shot: **Mistral-7B-Instruct-v0.3** (7B), **Meta-Llama-3-8B-Instruct** (8B), and **DeepSeek-R1-Distill-Llama-8B** (8B) — the backbone for FinMini.

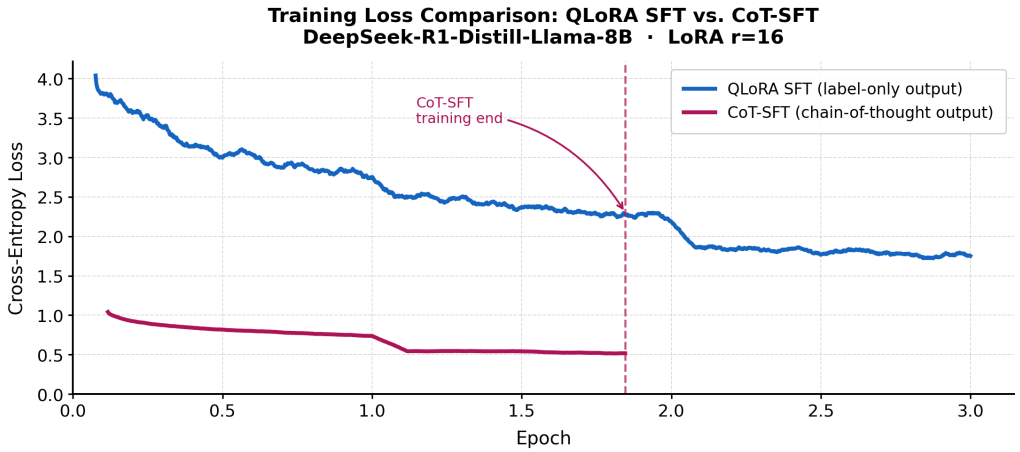

Figure 1: Training loss for QLoRA-SFT (blue) and CoT-SFT (pink) on DeepSeek-R1-Distill-Llama-8B with LoRA $r = 16$. The dashed line marks the end of the first stage. CoT-SFT converges to a lower loss owing to the richer chain-of-thought supervision signal.

Benchmark	Domain	Size	Description
FPB	EN News	393	Financial PhraseBank; partially in-distribution
FiQASA	EN Q&A	217	Financial Q&A opinions; unseen cross-domain
SM-BigData	Social	1,472	BigData22 Twitter stock movement; unseen
SM-CIKM	Social	1,143	CIKM18 Twitter stock movement; unseen

Table 2: Evaluation benchmarks from the Open FinLLM Leaderboard. FPB shares its source with the training FPB split; the remaining three benchmarks are entirely out-of-distribution.

4.3 Main Results

Figure 2 provides a visual comparison of all five models. FinMini CoT-SFT (pink bars, starred) achieves the highest accuracy on all four tasks, with dramatic improvements on formal financial benchmarks and consistent gains on social-media tasks.

4.4 Ablation: Contribution of Each Fine-Tuning Stage

Figure 3 visualizes the incremental accuracy gain from each fine-tuning stage relative to the DeepSeek-8B zero-shot baseline. Each bar is partitioned into three segments: the baseline accuracy (blue), the gain from QLoRA-SFT (amber), and the additional gain from CoT-SFT (rose). Dashed horizontal lines mark the baseline level for each benchmark.

QLoRA-SFT provides the dominant gain, particularly on FiQASA (+36.9 pp), where the baseline was clearly miscalibrated for Q&A-style financial sentiment. CoT-SFT adds a further +5.4 pp on FPB and +3.3 pp on SM-CIKM, confirming that structured reasoning supervision yields consistent gains beyond label-only training. Social-media benchmarks improve modestly at both stages, reflecting the limited coverage of informal financial language in the training corpus.

Model	Accuracy (%)				Avg
	FPB	FiQASA	SM-BD	SM-CIKM	
Mistral-7B (zero-shot)	85.5	72.8	49.1	47.3	63.7
Llama-3-8B (zero-shot)	69.5	82.0	43.1	39.3	58.5
DeepSeek-8B (zero-shot)	83.5	57.6	55.0	49.3	61.4
FinMini QLoRA-SFT	93.1	94.5	57.1	52.2	74.2
FinMini CoT-SFT (Ours)	**98.5**	**97.7**	**57.3**	**55.5**	**77.3**

Table 3: Accuracy on Open FinLLM Leaderboard benchmarks. Bold: best result per column. FinMini CoT-SFT outperforms all baselines on every benchmark.

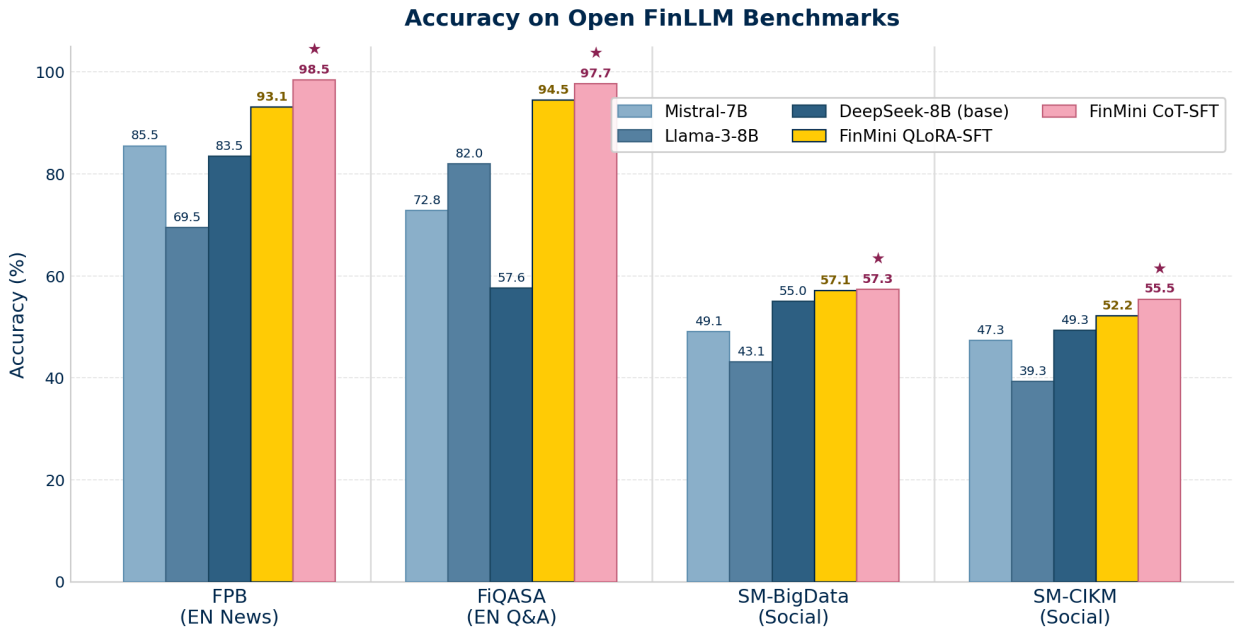

Figure 2: Accuracy on the Open FinLLM Leaderboard benchmarks. Stars (★) mark the best result per benchmark. FinMini CoT-SFT (pink) achieves the top score on all four tasks.

5 Analysis

5.1 Why Chain-of-Thought Supervision Helps

The gains from CoT-SFT are most pronounced on formal-language benchmarks (FPB: 93.1 → 98.5%, FiQASA: 94.5 → 97.7%). We attribute this to two complementary effects. **Format alignment**: the base model was distilled to produce `<think>` traces; QLoRA-SFT on label-only outputs partially suppresses this capability, whereas CoT-SFT reinforces it. **Reasoning quality**: structured traces force the model to identify domain-specific sentiment cues — hedged language in analyst commentary, implicit expectations in earnings guidance — before committing to a label, reducing ambiguity-driven errors.

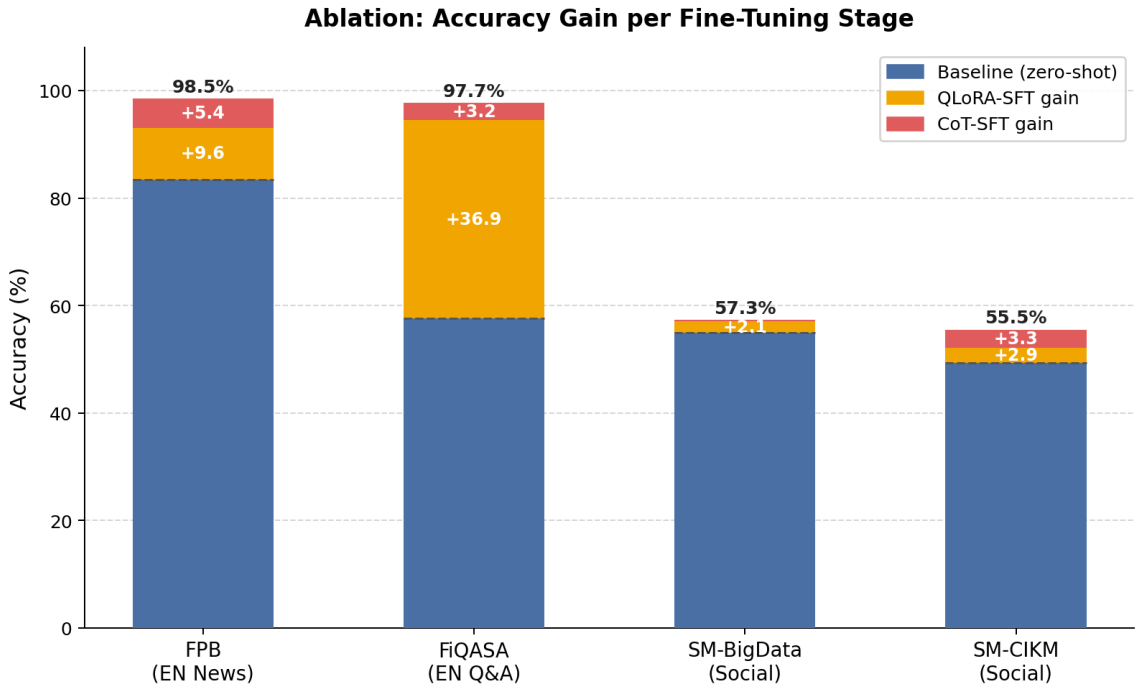

Figure 3: Stacked accuracy contributions per fine-tuning stage. Blue: DeepSeek-8B zero-shot baseline. Amber: gain from QLoRA-SFT. Rose: additional gain from CoT-SFT. Labels inside each segment show the per-stage accuracy increment (pp); bold labels above each bar show the final accuracy of FinMini CoT-SFT.

5.2 The Social-Media Performance Gap

All models exhibit a ∼30–40 pp accuracy gap between formal financial benchmarks and social-media tasks (Table 3). This is consistent with known challenges in social-media financial NLP [2, 4]: informal language, slang, hashtags, and ticker symbols are underrepresented in instruction-tuning corpora; sentences are shorter and more ambiguous; and class distributions differ substantially from news-style data. FinMini narrows the gap relative to all baselines — SM-BigData improves from 55.0% (baseline) to 57.3% (+2.3 pp), and SM-CIKM from 49.3% to 55.5% (+6.2 pp) — but does not eliminate it. Future work should incorporate Twitter and Reddit financial corpora directly into Stage 1 training.

5.3 LLM-Driven Long-Short Equity Strategy

To demonstrate a practical downstream application, we deploy FinMini CoT-SFT in a **quarterly long-short equity strategy** over the full S&P 500 universe. For each stock-quarter pair, a document is assembled from the earnings call transcript (fetched via the DefeatBeta API) and up to three recent news articles, truncated to 3,072 tokens. The model outputs a JSON assessment containing an `llm_score` (0–100), direction, confidence level, and a brief reasoning summary.

At each rebalance date (45 days after quarter-end), stocks are ranked cross-sectionally by `llm_score`. The top 10% are held long and the bottom 10% are held short, both equal-weighted. The experiment uses 12 signal quarters from 2023Q1 to 2025Q4, producing 11 realized quarterly holding-period returns.

Figure 4 shows the portfolio performance. The strategy achieves 21.83% cumulative return, 7.44% annualized return, Sharpe 0.69, and -9.47% max drawdown. The long-short portfolio grows from 1.00 to 1.22. The long leg grows to 1.89, while the low-score basket also rises to 1.54, which reduces the final spread. The strongest quarter is 2024Q2, with an 11.21% long-short return; the weakest quarter is 2025Q3, with an −8.38% return.

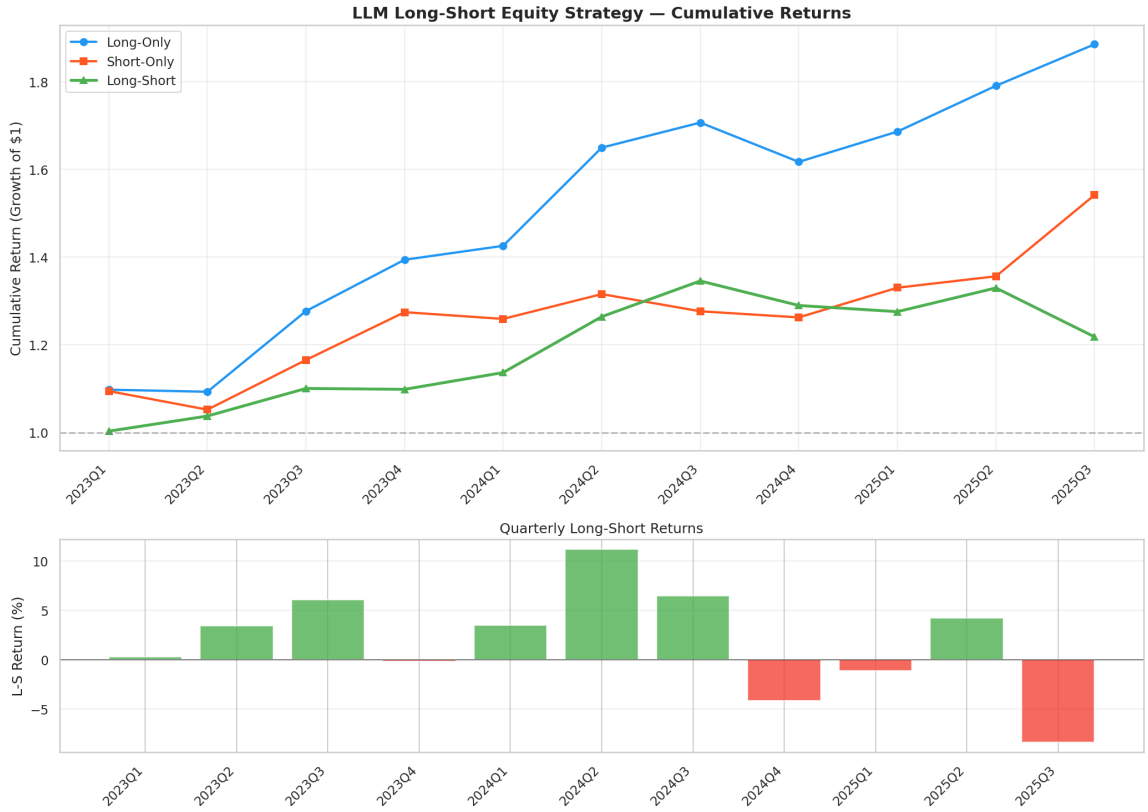

Figure 4: Cumulative returns and quarterly long-short returns for the LLM-driven equity strategy.

The reasoning cases also match the portfolio intuition. High-score stocks are usually linked to strong earnings, raised guidance, AI-driven growth, or confident management language (e.g., GOOGL, ADBE, AMD, and ABBV). Low-score stocks often have missing transcripts, weak guidance, or clear negative news signals (e.g., ALB and KVUE). Overall, this application shows that FinMini's text scores can provide a usable cross-sectional signal, though the short leg remains noisy in a rising market.

6 Conclusion

FinMini shows that a compact 8B reasoning-distilled LLM can be effectively adapted to financial NLP through QLoRA-SFT and CoT-SFT. It achieves the best accuracy among all tested baselines across four Open FinLLM benchmarks, with especially strong results on formal financial text. The long-short equity strategy further shows that FinMini's LLM-derived scores can serve as a useful downstream signal for portfolio construction, although the short leg remains noisy in a rising market.

Author Contributions

JiaYuan Jiang: *Optimized and deployed the training pipeline on the Great Lakes HPC cluster, including checkpoint/resume support and hyperparameter tuning. Analyzed training and benchmark results, designed the conference poster, and authored the final report.*

JiaYi Zhang: *Proposed the core project concept and implemented the main SFT and CoT-SFT training code. Designed and implemented the LLM long-short strategy pipeline, including data collection, LLM scoring, backtesting, visualization, and result analysis.*

Yue Yu: *Built the data pipeline, including dataset selection, preprocessing, label normalization, instruction formatting, CoT data augmentation, and benchmark preparation. Co-authored the final project report.*

YiXiang Fei: *Contributed to project ideation, task formulation, dataset and evaluation discussions, and the initial poster design. Reviewed the final report for organization, clarity, and consistency with the poster.*

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